EQUALITY IN THE POLAR SCHNEIDER COLLECTION INEQUALITY

ARXIV_AUTOMATIC_MATH RUN FA_BANACH_001

Source and Status

This packet concerns Julian Haddad, Dylan Langharst, Galyna V. Livshyts, and Eli Putterman, *On the polar of Schneider's difference body*, arXiv:2503.06191.

Status: `full_solution_likely_valid` for the equality-condition gap in the source paper's Theorem 11.

Source Question

For a collection $\mathcal{K} = (K_0, \dots, K_m)$ of convex bodies in $\mathbb{R}^n$, the source defines

$$D^m(\mathcal{K}) = \{(x_1, \dots, x_m) : K_0 \cap (K_1 + x_1) \cap \cdots \cap (K_m + x_m) \neq \varnothing\}.$$

Theorem 11 of the source paper states that, if all K_i have center of mass at the origin and $|K_i^\circ|_n = |K_0^\circ|_n$, then

$$|D^{m,\circ}(\mathcal{K})|_{nm} \prod_{i=1}^m |K_i|_n \leq |B_2^n|_n^m \, |D^{m,\circ}(B_2^n)|_{nm}.$$

It further states that equality is necessary only if each K_i is a centered ellipsoid, and sufficient if all K_i are the same centered ellipsoid. The authors explicitly note a gap: they do not know whether equality can occur when all K_i are centered ellipsoids but at least one differs from the others.

Idea of the Proof

The only missing case is a tuple of centered ellipsoids. Write

$$h_{E_i}(u)^2 = u^T Q_i u$$

with Q_i positive definite. Equal polar volumes are exactly the condition that $\det Q_i$ is independent of i. The polar volume of $D^m(E_0, \dots, E_m)$ is an integral of $\exp(-h_{D^m(E_0,\dots,E_m)})$. The elementary Laplace representation of $e^{-\sqrt{s}}$ writes each factor as a positive mixture of Gaussians.

For each fixed set of mixture parameters $t_0, \dots, t_m > 0$, the inner Gaussian integral is controlled by the determinant of a block matrix. The source paper already computes this determinant in its functional section:

$$\det M = \prod_{i=0}^m \det(t_i Q_i) \, \det\left(\sum_{i=0}^m (t_i Q_i)^{-1}\right).$$

Minkowski's determinant inequality gives a pointwise lower bound for this determinant, with strict equality unless all Q_i are equal. Since the mixture is positive, the strict pointwise inequality integrates to strict volume loss for every non-identical ellipsoid tuple.

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Main Theorem

Theorem 1. *Let $m, n \in \mathbb{N}$. Under the hypotheses of Theorem 11 of Haddad, Langharst, Livshyts, and Putterman, equality holds if and only if*

$$K_0 = K_1 = \cdots = K_m$$

is a centered ellipsoid.

Proof. The source paper already proves the inequality, proves equality for identical centered ellipsoids, and proves that equality can occur only when every K_i is a centered ellipsoid. It remains to rule out equality for different centered ellipsoids.

Let $E_i = A_i B_2^n$ be centered ellipsoids and put $Q_i = A_i A_i^T$, so

$$h_{E_i}(u) = \sqrt{u^T Q_i u}.$$

The condition $|E_i^\circ|_n = |E_0^\circ|_n$ is equivalent to

$$\det Q_i = \det Q_0 =: \delta \qquad (0 \le i \le m).$$

Indeed $|E_i^\circ|_n = |B_2^n|_n / \sqrt{\det Q_i}$.

For $z = (z_1, \dots, z_m) \in (\mathbb{R}^n)^m$, the support-function formula in the source paper gives

$$h_{D^m(E_0,\dots,E_m)}(z) = \sqrt{\left(\sum_{j=1}^m z_j\right)^T Q_0 \left(\sum_{j=1}^m z_j\right)} + \sum_{i=1}^m \sqrt{z_i^T Q_i z_i}.$$

Using the standard polar-volume formula in dimension $N = nm$,

$$|D^{m,\circ}(E_0, \dots, E_m)|_{nm} = \frac{1}{(nm)!} \int_{(\mathbb{R}^n)^m} e^{-h_{D^m(E_0,\dots,E_m)}(z)}\, dz.$$

We use the positive Gaussian-mixture identity

$$e^{-r} = \frac{1}{\sqrt{2\pi}} \int_0^\infty t^{-3/2} \exp\left[-\frac{1}{2}\left(tr^2 + t^{-1}\right)\right] dt \qquad (r \ge 0).$$

Applying this identity to the $m+1$ square-root terms in the support function and using Fubini, the volume integral is a positive average, over $t_0, \dots, t_m > 0$, of Gaussian integrals of the form

$$\int_{(\mathbb{R}^n)^m} \exp\left[-\frac{1}{2}\left(t_0 \left(\sum_{j=1}^m z_j\right)^T Q_0 \left(\sum_{j=1}^m z_j\right) + \sum_{i=1}^m t_i z_i^T Q_i z_i\right)\right] dz.$$

The corresponding block matrix $M(t,Q)$ has diagonal blocks $t_iQ_i + t_0Q_0$ and off-diagonal blocks t_0Q_0. By the determinant calculation in Proposition 30 of the source paper, applied to the matrices t_iQ_i,

$$\det M(t,Q) = \prod_{i=0}^m \det(t_iQ_i)\, \det\left(\sum_{i=0}^m (t_iQ_i)^{-1}\right).$$

The inner Gaussian integral is $(2\pi)^{nm/2} \det M(t,Q)^{-1/2}$.

By Minkowski's determinant inequality for positive-definite matrices,

$$\det\left(\sum_{i=0}^m t_i^{-1} Q_i^{-1}\right)^{1/n} \ge \sum_{i=0}^m t_i^{-1} \det(Q_i^{-1})^{1/n} = \delta^{-1/n} \sum_{i=0}^m t_i^{-1}.$$

Equivalently,

$$\det\left(\sum_{i=0}^{m}(t_iQ_i)^{-1}\right) \geq \delta^{-1}\left(\sum_{i=0}^{m} t_i^{-1}\right)^n.$$

Therefore

$$\det M(t,Q) \geq \delta^m \left(\prod_{i=0}^{m} t_i^n\right)\left(\sum_{i=0}^{m} t_i^{-1}\right)^n.$$

This lower bound is exactly the value of $\det M(t,Q)$ when all Q_i are replaced by a single positive-definite matrix Q with $\det Q = \delta$.

The equality condition in Minkowski's determinant inequality says that equality holds only when the matrices Q_i^{-1} are pairwise proportional. Since all $\det Q_i$ are equal, this means $Q_0 = \cdots = Q_m$. Hence, if the ellipsoid tuple is not identical, then for every $t_i > 0$,

$$\det M(t,Q) > \delta^m \left(\prod_{i=0}^{m} t_i^n\right)\left(\sum_{i=0}^{m} t_i^{-1}\right)^n,$$

and the corresponding Gaussian integral is strictly smaller than for the identical ellipsoid tuple. Since the mixing measure above is positive, the strict inequality survives integration. Thus

$$|D^{m,\circ}(E_0,\ldots,E_m)|_{nm} < |D^{m,\circ}(E,\ldots,E)|_{nm}$$

whenever the E_i are not all equal and E is any centered ellipsoid with $\det(E) = \det(E_i)$.

Finally,

$$\prod_{i=1}^{m}|E_i|_n = |E|_n^m,$$

so a non-identical ellipsoid tuple gives a strictly smaller left-hand side than the identical ellipsoid tuple. The identical tuple is an equality case by the source paper. Therefore equality in Theorem 11 occurs exactly for identical centered ellipsoids. □

Corollary 1. *The earlier $m = 1$ partial packet is a special case: if E_0, E_1 are centered ellipsoids with $|E_0^\circ|_n = |E_1^\circ|_n$, equality in the $m = 1$ collection inequality holds if and only if $E_0 = E_1$.*

Limitations

This solves the equality gap in the source paper's polar Schneider collection inequality. It does not solve Schneider's original higher-order conjecture that $|D^m(K)|_{nm}|K|_n^{-m}$ is minimized by ellipsoids for $n \geq 3$ and $m \geq 2$.

Novelty Check

Local run indexes were searched for `2503.06191`, "Polar Schneider", "Theorem 11", "collection equality", "same centered ellipsoid", and "Rogers-Brascamp-Lieb-Luttinger equality". The only local packet was the earlier $m = 1$ partial result, now superseded by this packet.

On 2026-07-03, bounded web searches for the exact title and phrases including "Polar Schneider inequality equality collection ellipsoids", "same centered ellipsoid", and "Rogers-Brascamp-Lieb-Luttinger Schneider equality" surfaced the source arXiv page and no separate answer to the equality gap. Novelty confidence is moderate: the proof uses elementary tools and a determinant calculation already present in the source paper's functional section, so the observation may be accessible to the authors or experts.

Human Review Recommendation

Review as a likely valid full solution to the source equality-condition gap. The key checks are:

- the positive Gaussian-mixture identity for e^{-r};
- the determinant formula for the block Gaussian matrix after replacing the source matrices by t_iQ_i;
- the strict equality condition in Minkowski's determinant inequality.

References

[1] J. Haddad, D. Langharst, G. V. Livshyts, and E. Putterman, *On the polar of Schneider's difference body*, arXiv:2503.06191, 2025.

[2] S. Artstein-Avidan, A. Giannopoulos, and V. D. Milman, *Asymptotic Geometric Analysis, Part I*, Mathematical Surveys and Monographs 202, American Mathematical Society, 2015.